

World Happiness Report Dataset

Introduction

The dataset I used for this project is the World Happiness Report. The data is collected by Gallup World Poll, a global survey which spans 140+ countries which they claim to cover 95% of global adults, the survey is conducted via face-to-face interviews or telephone. more than 100,000 people are surveyed each year, but only about 1000 people per country each year. The timing, techniques, and sampling locations are carried out to account for error. e.g. Gallup accounts for seasonal changes, war, and other factors which might contribute to random error. The "Life evaluation (3-year average)" column is measured in a universal fashion. Not every country has a scale from 1 to 10 funnily enough. In the dataset, the 6 factors and "Dystopia + Residual" columns were added starting in 2019 after they concluded they somewhat correlate with happiness rather than causing it. Unemployment and equality are not accounted for because they aren't available yet for the full sample of countries and they ostensibly have a weaker correlation with happiness than the other measures. All these methods, using three-year averages, accounting for cultural and regional differences all sound accurate.

Limitations

There are several limitations which may make WHR unrepresentative of happiness across the world, however. As of 2026, there are a total of 195 recognised countries in the world. And the WHR has only included 147 of those countries in the latest findings (in 2025). They mention that they don't survey small or inaccessible populations due to it being cost-ineffective to travel to. This could bring up many questions. Like, at what point would the state be accessible, and hence cost-effective to travel to? What is the minimum population they are willing to assess? Depending on their answers, specific states will be completely unaccounted for which skews the rankings with states with higher populations. Another possible limitation of the WHR would be subjectivity within the measurements themselves. Although the method they use to gather results for "Life evaluation (3-year average)" is universal, countries will still see them completely different. Like in western culture, a 10/10 is seen as perfection so it is rarely given out. 8-9 is considered high praise. But in eastern culture, they may prioritise collective thinking rather than individualism. Because of this, ratings may converge to 8-9, or skew downwards, or any combination of data. Even if these limitations are unaccounted for, the WHR still seems like accurate representation of worldwide happiness, as each country has some context for their individual rank, like Finland being ~0.2 above 2nd place almost every year, they seem to be doing the right things.

Assumptions

There were a few assumptions that I have made about the dataset.

- the "Life evaluation (3-year average)" column is representative of happiness - the happiness ranking is based on life evaluation, a question derived from the Cantril ladder.

- 5 entries across a span of 5 years is enough to filter out anomalous observations - after seeing a few outliers with only 2 to 4 entries, I decided to remove them as they would not be ranked fairly due to the other countries having at least 20+ entries.

Cleaning

During the cleaning process, I didn't really manipulate the data much. I did remove the upper and lower bound of life_eval as I felt that the margin of error wasn't needed in this analysis due to the focus on correlations. Other than that, removing empty columns and renaming them for readability made it easier for me to analyse.

Results

I aimed to answer these three questions during my analysis:

1. Does money (GDP) correlate with happiness? With high-income countries, is the relationship linear or diminishing?
2. Does generosity correlate with happiness? Does it remain significant when accounting for other social, economic factors?
3. What common characteristics define countries that consistently improve in happiness rankings over time, and what factors might explain these trends?

The results prove to be interesting. Log (GDP) had a moderate positive correlation with happiness (0.682). As expected, having more money is correlated to having more happiness, although, log (GDP) is used so we can't conclude that the relationship is diminishing. Generosity has no correlation with happiness, removing generosity resulted in a small decrease in model performance by -0.019, and generosity yielded the lowest coefficient when standardising for all features, which all surprisingly enough, point to generosity being insignificant in context to happiness. social support (having someone to talk to) seems to be correlated with top improvers in happiness, freedom to make life choices sees small increases in improvers, and the other factors seem to be uncorrelated. Notably, GDP isn't a key characteristic in improvers. So, the only telling characteristic when it comes to improvers is having someone to talk to during rough times. Who knew human connection is what we needed!

Overall, this analysis has been very intriguing. We will never be able to quantify human emotion in data and statistics but discovering that social support being correlated with happier countries, and generosity being uncorrelated was very surprising. Working through this dataset has given me a wider perspective when looking at happiness in different countries, especially in higher-income countries. Looking at the stories behind the decliners was also insightful. And I am eager for the WHR 2026 data!